



Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

CSC1108 Final Year Project Proposal

BSc in Data Science

Phenotyping with AI Agents

Sarah O'Hanlon (22444924)

Tanuli Liyanage (22100555)

Supervised by

Tomas Ward

Student emails

sarah.ohanlon24@mail.dcu.ie

tanuli.damburaliyanage2@mail.dcu.ie

Executive Summary

This project investigates how different large language models (LLMs) and humans learn and make decisions under uncertainty. Traditional AI evaluation focuses on accuracy or reward, but these metrics do not reveal the underlying behavioural strategies that agents use when interacting with different environments. Inspired by phenotyping in psychology, this study aims to characterise the behavioural profiles of agents by examining how they adapt, explore and respond during cognitive decision making tasks.

We propose to use the two armed bandit game, a simple yet powerful reinforcement learning framework widely used in human cognition research [12]. LLMs will be prompted to act as decision making agents, while behavioural baselines will be drawn from previously collected human two armed bandit datasets. Each agent’s sequence of choices and rewards will be modelled using the Rescorla–Wagner learning rule, allowing us to estimate key behavioural parameters such as learning rate, reward sensitivity, and exploration tendency.

By mapping these parameters into a behavioural space and clustering the resulting agent profiles, we aim to identify distinct behavioural phenotypes and compare how closely LLM decision patterns align with those observed in human participants. This work contributes to a deeper, data-driven understanding of AI behaviour, offering insights into interpretability, predictability, and the psychological plausibility of emerging AI systems.

1. Motivation and Background

Artificial intelligence systems increasingly make autonomous decisions in environments that require learning, adaptation, and interaction. Yet, despite dramatic advances in model performance, from reinforcement-learning agents mastering control tasks to large language models (LLMs) reasoning through natural language, the field still lacks a clear understanding of how and why these systems behave the way they do. We can measure accuracy or total reward, but such metrics conceal the fact that two agents achieving similar scores may rely on entirely different decision-making strategies.

In biological and psychological sciences, analysing behavioural variability is achieved through phenotyping, which refers to the ‘observable types of an organism, and it can refer to anything from a common trait, such as height or hair color, to presence or absence of a disease’ [21]. Translating this idea to artificial systems opens a new way of studying machine behaviour: rather than judging models solely by performance, we can describe their behavioural profiles. For instance, one reinforcement learning agent may learn quickly but explore little, while another learns slowly yet remains flexible. Understanding such differences could inform the design of more predictable, interpretable, and trustworthy AI systems.

From a data-science perspective, the challenge is both analytical and methodological. Learning behaviour emerges from the interaction of model parameters (learning rate, exploration tendency, reward sensitivity) and environmental uncertainty. Mapping this interaction produces a high dimensional behavioural space in which each agent represents a unique point. Analysing and clustering these data can reveal distinct behavioural types,

analogous to cognitive or personality profiles in humans. Doing so requires combining quantitative modelling, simulation, and exploratory data analysis, core strengths of data science.

The importance of this research lies in its potential to make AI behaviour more transparent. As machine-learning systems are deployed in decision critical contexts, understanding the range and stability of their behaviours becomes essential for safety, reproducibility, and alignment with human expectations. Insights from such controlled environments can inform broader questions about interpretability and model diversity. Progress on this problem would not only advance fundamental understanding of learning dynamics but also contribute practical methods for evaluating and comparing autonomous systems in terms of how they act, not merely how well they perform.

2. The Problem Statement

Understanding how different intelligent agents learn and make decisions under uncertainty is a central question in artificial intelligence, psychology and computational psychiatry. The success of AI systems has been immense, however the capacity of current computational technologies in psychiatry has reached a stage where its integration into psychiatric practice is not just feasible but urgently needed. And the remaining obstacles that now need to be addressed are no longer rooted in technological advancements but in ethics, education and understanding [17]. Unlike traditional reinforcement learning algorithms, LLMs are not explicitly trained to optimise reward in sequential environments, yet they are increasingly being used in settings that require decision making [13, 14]. This raises the fundamental question:

Do LLMs exhibit consistent behavioural phenotypes when placed in a cognitive decision making task, and how do these patterns compare across different agents and does it align just as well with humans?

Therefore, our objective in this study is to apply the concept of phenotyping to identify behavioural phenotypes that emerge from different large language models (LLMs) during a cognitive decision making task and compare these behaviours to human participants who were placed in the same environment.

Specifically, we will study a two-armed bandit task, a reinforcement learning environment where an agent repeatedly chooses between two uncertain rewards. Each agent's gameplay will generate a unique dataset capturing its choices, rewards received and adaptation patterns. We then fit a classical Rescorla Wagner reinforcement learning model to each agent's decisions to quantify their behavioural differences in learning rate, reward sensitivity and observe the exploration and exploitation tendencies. Our work will focus on analysing two main categories of agents:

1. Large language models (eg: ChatGPT, Gemini, Claude, DeepSeek), interacting via their APIs and prompted to act as decision making agents across repeated trials.
2. Human participants, whose gameplay will provide a natural behavioural baseline for comparison.

The scope of this project is general and exploratory. Our ultimate goal is not to optimise the performance of AI agents or to build a new decision making model but to phenotype and compare decision making behaviours of LLMs and humans using a common cognitive framework. By doing so, we intend to contribute to a data driven understanding of LLM behaviour and explore whether they mirror similar behavioural patterns to humans in ways relevant to psychology and computational psychiatry.

3. State of the Art

Understanding how artificial agents behave has become an increasingly important topic across artificial intelligence, cognitive science, and data science. As AI systems grow more capable, researchers have begun to look beyond accuracy and reward-based metrics to examine the behavioural patterns that emerge when these systems interact with structured tasks. This shift is captured clearly in work by Binz and Schulz (2023), who argue that performance scores alone obscure the underlying decision-making processes of large language models. By treating GPT-3 as a participant in controlled cognitive psychology experiments, they demonstrate that behavioural signatures, such as tendencies in reasoning, exploration or causal judgement, offer deeper insight into how AI systems operate than correctness metrics alone [1]. This critique aligns with Bender and Koller (2020), who argue that high performance on benchmark tasks does not imply genuine understanding or robust decision-making capacity. Their work highlights the limitations of assessing language models solely by accuracy, reinforcing the need for behavioural analysis to uncover how these systems arrive at their outputs [11]. These papers highlight a wider recognition that analysing how AI systems make decisions provides deeper insight into their capabilities and limitations than evaluating whether their answers are correct.

This perspective has led to a convergence of methods from cognitive psychology, reinforcement learning, and interpretability, creating a research area in which AI models are studied much like human participants in behavioural experiments. Rather than asking only whether a model arrives at the right answer, researchers increasingly examine how it arrives there, how it explores uncertain choices, how it responds to risk or ambiguity, how its behaviour changes under different contextual cues and whether it exhibits stable, trait-like patterns across tasks. These approaches highlight that AI systems can display consistent behavioural tendencies or “phenotypes” that emerge from their learning dynamics, decision-making structures, or even the prompts they receive.

Studies such as [2], reveal that large language models can exhibit stable behavioural patterns in emotional contexts, performing and even generating emotional intelligence tests at human-like level. Their findings underline the viability of behavioural frameworks for analysing AI agents beyond raw performance metrics. While their focus is on socio-emotional tasks, the underlying principle of mapping agent behaviour in controlled tasks is directly relevant to this project, which applies a simplified two armed bandit paradigm to explore phenotypes of artificial learners.

In parallel, research has shown that LLM behaviour can be systematically manipulated through prompt engineering. By instructing a model to answer “as an anxious person,” “as a

confident decision-maker,” or “as someone with depressive symptoms,” researchers have demonstrated stable shifts in reasoning style, emotional tone and risk preferences [3]. These results indicate that AI systems exhibit behavioural malleability, where distinct decision profiles can be induced through external instructions. This parallels the idea that behavioural phenotypes in artificial agents may arise from interactions between model architecture, input framing and internal learning dynamics.

In cognitive science, the multi-armed bandit task has long served as a core paradigm for studying how humans balance exploration and exploitation when faced with uncertain rewards. Speekenbrink and Konstantinidis (2015) show that even in more complex “restless” bandit environments, where reward probabilities change over time, humans display systematic strategies shaped by uncertainty, risk sensitivity, and learning dynamics. Their findings demonstrate that bandit tasks reveal meaningful variation in decision-making behaviour, making them a powerful tool for behavioural analysis [10]. This psychological foundation supports the use of the two armed bandit in AI research, if the task can uncover distinct behavioural patterns in humans, it provides an equally controlled and interpretable framework for examining behavioural differences in artificial agents.

Another key direction examines AI behaviour using structured decision-making tasks traditionally used in psychology and neuroscience. For decades, the two armed bandit has been a central paradigm for studying human exploration-exploitation trade-offs, revealing how individuals balance uncertain rewards. This paradigm is now being repurposed to analyse AI models. Zhang et al. (2025) delved deep into this topic and their results reveal that different LLMs exhibit distinct behavioural profiles, with some showing over-exploitation (sticking to one choice), while others demonstrate erratic exploration or susceptibility to adversarial manipulation. These patterns highlight model-specific vulnerabilities in strategic reasoning and adaptability. The authors argue that, rather than relying solely on accuracy benchmarks, analysing behavioural dynamics in such controlled decision-making environments provides a more meaningful lens for assessing model robustness and cognitive alignment. This research is directly relevant to the present project, which extends similar behavioural analysis principles to simpler reinforcement learning agents. By using the two armed bandit environment and varying internal parameters such as learning rate and exploration tendency, the current work aims to produce phenotypes of artificial agents analogous to those behavioural differences observed in LLMs [4].

Beyond large language models, reinforcement learning research provides foundational insight into how behavioural differences emerge from variations in learning mechanisms. Classic models such as the Rescorla–Wagner rule [5] show that even simple learning systems display distinct behaviour depending on factors such as learning rate or reward sensitivity. In cognitive and clinical modelling, parameter variation has been used to characterise human learning styles and disorders, illustrating how formal models capture behavioural phenotypes [6]. Within artificial agents, researchers have developed methods for representing and comparing entire policies rather than focusing solely on returns. Within artificial agents, researchers have developed methods for representing and comparing entire policies rather than focusing solely on returns. The most prominent example is the concept of behaviour space, introduced by Stork et al. (2020). By analysing Q-learning agents in deterministic maze and continuous inverted pendulum environments, they show that agents achieving similar rewards may follow entirely different action trajectories. Mapping agents into a behaviour space enables clustering,

visualisation and systematic comparison of strategy types, forming a methodological foundation for phenotyping work [7].

In addition to behaviour-space representations, interpretability research in reinforcement learning has developed complementary methods for analysing agent decision policies. Amado et al. (2020) provide a comprehensive survey of interpretation techniques for RL systems, highlighting approaches such as policy summarisation, saliency-based state attribution, reward decomposition, and model distillation into human-readable forms. These methods share the underlying goal of explaining how an agent arrives at its decisions rather than evaluating only its final performance. Importantly, the survey emphasises that RL agents often exhibit subtle behavioural differences that are not reflected in reward metrics alone, necessitating tools that expose patterns in exploration, action selection, and state sensitivity. This aligns closely with the aims of the present project, which seeks to characterise behavioural phenotypes by identifying stable patterns that emerge when learning parameters vary. While Amado et al. focus on interpretability at the policy or state-importance level, the behaviour clustering approach adopted in this project provides a complementary perspective by examining emergent behavioural structure across simulated agents [8].

Beyond the methodological need to analyse agent behaviour, research in AI safety highlights the broader importance of understanding behavioural variability in autonomous systems. Amodei et al. (2016) identify several “concrete problems” in AI safety, such as reward hacking, unexpected side effects, and unsafe exploration, that arise precisely because agents may behave in ways that designers did not anticipate. Their analysis emphasises that even small changes to an agent’s objectives, environment, or learning parameters can lead to large and sometimes undesirable behavioural shifts. This perspective reinforces the value of systematically mapping behavioural tendencies rather than relying solely on aggregate reward performance [9]. In this sense, behavioural phenotyping offers a structured approach to identifying how and why different learning configurations produce distinct patterns of action, providing insights that are directly relevant to preventing unintended behaviours in deployed AI systems.

4. Methodology

4.1 Overview of the Approach

This project adopts a behavioural analysis approach within a reinforcement learning environment. Our main goal is to investigate how different agents (LLMs and humans) make sequential decisions to maximise reward in a two-armed bandit task under uncertainty. Each agent will be prompted to interact with the same environment where two possible actions result in probabilistic rewards. By logging every action, reward and outcome, we will produce comparable datasets that will then be used to model each agent’s behaviour using the Rescorla Wagner learning rule to analyse their behavioural differences.

4.2 Experimental Environment: The Two-Armed Bandit Task

The two-armed bandit is a classical reinforcement learning environment where a decision

maker consistently chooses between two alternative arms over a sequence of trials to better understand and maximise the total number of rewards [15]. Each arm has a fixed but unknown probability of receiving a reward and at each of the n stages the result of the pull is known to the player before an arm is pulled at the next stage. When choosing an arm, the decision-maker must balance exploitation (choosing the current best option) with exploration (choosing an uncertain option with the potential for greater long term success) as future gains are as valuable as the immediate ones [16].

We will be implementing our two-armed bandit environment using Python and the environment will generate reward outcomes based on fixed probabilities (eg: Arm A=0.6, Arm B=0.4). This environment will serve as the identical testing ground for all AI agents and all human participants throughout our study to ensure comparability.

4.3 AI Agents and Human Data

A key novelty of our project is interfacing LLMs with a reinforcement learning environment in a way that forces them to behave like decision-making agents. To make our phenotyping analysis more meaningful, we will use the APIs of multiple LLMs from different families such as:

- ChatGPT 5.1 (OpenAI)
- Gemini 1.5 (Google)
- DeepSeek R1
- Claude 3.5

Each LLM will receive a structured prompts such as:

“You are playing a two-armed bandit game. Your task is to choose between Arm A or Arm B. Your goal is to maximise reward. The probability of getting a reward on each arm is unknown at the start but you can learn and adjust your strategy based on the outcomes of your previous visits. Respond only with one word: “A” for Arm A or “B” for Arm B.”

At every trial, along with the above prompt, the LLM will receive the trials so far, the options chosen and feedback on their decision, either recording a reward or failure. The LLM then outputs the next action. This process results in a behavioural dataset for each AI agent that logs all the responses and rewards which will be stored for analysis.

Our analysis aims to understand whether LLMs display human-like, subhuman, or superhuman decision-making behaviours. Human gameplay data will come from an existing public dataset employed by Zhang et al. (2025) in their research on adversarial evaluation framework to diagnose behavioural weakness of LLMs [4].

4.4 Behaviour Modelling Using the Rescorla-Wagner Model

The Rescorla Wagner (RW) model is a foundational theory of classical conditioning designed to explain how learning occurs on a trial by trial basis [18].

In the RW model, learning occurs through quantitative predictions, given three parameters: a participant's prior knowledge of an association and salience of both conditioned stimuli (CS) and unconditioned stimuli (US). The amount of learning that occurs in one trial will depend on how surprised the player is by the unconditioned stimulus. The level of surprise is computationally defined as the difference between the maximum possible associative strength achievable for a given CS - US pairing (λ , or lambda) and the current total learned association (V_T). V_t represents the learner's current expectation of the unconditioned stimuli based on all conditioned stimuli present [19].

The overall rate of learning is controlled by the salience of both the CS (α , or alpha) and the US (β , or beta) where the stimuli with higher salience lead to faster learning. Hence, the greatest learning occurs on early trials as the difference between the maximum possible learning (λ) and the current expectation (V_T) is at its largest. Using the equation below, the model formalises the change in associative value (ΔV) on each and every trial:

$$\Delta V = \alpha\beta(\lambda - V_T)$$

In our project, the Rescorla Wagner model is used to model participants choice behaviour in the two-armed bandit task and by fitting the model to the decisions of LLMs and humans, we can estimate their learning rates and compare how each group uses feedback to guide future choices. For example, a high learning rate indicates a highly reactive learner while a low learning rate suggests a slower and a more conservative learner. Comparing these will allow us to quantify similarities and differences in the underlying learning mechanisms of each agent and their decision making strategies.

4.5 Software Tools and Evaluation Procedure

Primary tools we will utilise throughout our project are:

- Python
- Jupyter Notebooks
- NumPy, Pandas
- SciPy (for RW model optimisation)
- Matplotlib/Seaborn
- OpenAI API / Gemini API / DeepSeek API
- GitHub for version control

We aim to evaluate our results by comparing the learning dynamics of human participants and LLM based agents within the two-armed bandit task. The main way we plan to do this is by looking at the parameters we get after fitting the RW model to each agent's choices. These parameters include learning rate, bias and noise. We will then compare these parameters between humans and LLMs using simple statistical tests such as t-tests or ANOVA to see if these two groups behave similarly or differently.

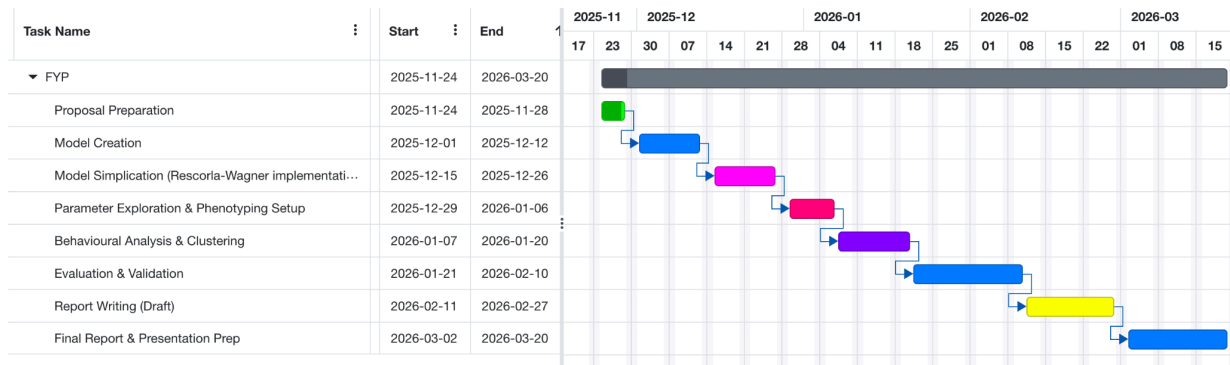
Additionally, we plan to use clustering methods like K-Means or Principal Component Analysis (PCA) to the RW parameters to see whether humans and LLMs form similar behavioral groups.

We would also check how well the RW model fit each agent’s behaviour by calculating the root mean squared error (RMSE) between the model’s predictions and the actual choices. A lower error would mean the model explains the behaviour well while a higher error could suggest that the agent is behaving in an inconsistent way [20]

In addition we assess the stability of each agent’s behaviour across trials by observing whether their decision strategies remain consistent or fluctuate over time. Consistent behaviour would suggest clear learning while large fluctuations may show that the agent is not learning effectively.

Finally we plan to carry out a qualitative analysis of the LLMs prompts to understand the aspects of their behaviour such as whether the LLM appears cautious, risk seeking, confident or random.

5. Project Plan and Gantt chart



The project is structured into sequential phases running from late Semester 1 into Semester 2. The above image shows the Gantt chart outlining all major tasks, their duration, and dependencies. The plan ensures that model development, experimentation, analysis, and writing are balanced across the semester and that work intensifies at appropriate points leading up to submission.

The project begins with proposal preparation (24–28 Nov 2025), followed by an initial model creation phase (1–12 Dec 2025) to implement the basic two-armed bandit environment and agent–LLM interaction pipeline. This is followed by model simplification and Rescorla–Wagner integration (15–26 Dec 2025), ensuring that the cognitive-style learning model is operational before the main experimental phase.

Early Semester 2 focuses on parameter exploration and phenotyping setup (29 Dec 2025 – 6 Jan 2026), preparing the analysis framework and identifying behavioural metrics. This leads into behavioural analysis and clustering (7–20 Jan 2026), where agent trajectories are modelled and embedded into a behavioural space.

The evaluation and validation stage (21 Jan – 10 Feb 2026) examines stability, cross-agent comparison, and differences between LLMs and existing human data. The remainder of Semester 2 is dedicated to documentation: draft report writing (11–27 Feb 2026) and final report and presentation preparation (2–20 Mar 2026).

References

- [1] M. Binz and E. Schulz, “Using cognitive psychology to understand GPT-3,” *Proceedings of the National Academy of Sciences*, vol. 120, no. 6, Feb. 2023, doi: <https://doi.org/10.1073/pnas.2218523120>.
- [2] K. Schlegel, N. R. Sommer, and M. Mortillaro, “Large language models are proficient in solving and creating emotional intelligence tests,” *Communications Psychology*, vol. 3, no. 1, May 2025, doi: <https://doi.org/10.1038/s44271-025-00258-x>.
- [3] Y. Wang, J. Zhao, D. S. Ones, L. He, and X. Xu, “Evaluating the ability of large language models to emulate personality,” *Scientific Reports*, vol. 15, no. 1, Jan. 2025, doi: <https://doi.org/10.1038/s41598-024-84109-5>.
- [4] L. Zhang, H. Wang, L. Cheng, L. Deng, and T. Ward, “Adversarial Testing in LLMs: Insights into Decision-Making Vulnerabilities,” *arXiv.org*, 2025. <https://arxiv.org/abs/2505.13195>.
- [5] . Rescorla, “Rescorla-Wagner Model,” *Scholarpedia*, vol. 3, no. 3, p. 2237, 2008, doi: <https://doi.org/10.4249/scholarpedia.2237>.
- [6] A. Dezfouli, K. Griffiths, F. Ramos, P. Dayan, and B. W. Balleine, “Models that learn how humans learn: The case of decision-making and its disorders,” *PLOS Computational Biology*, vol. 15, no. 6, p. e1006903, Jun. 2019, doi: <https://doi.org/10.1371/journal.pcbi.1006903>.
- [7] J. Stork, M. Zaefferer, T. Bartz-Beielstein, and A. E. Eiben, “Understanding the Behavior of Reinforcement Learning Agents,” May 06, 2020. https://www.researchgate.net/publication/341286142_Understanding_the_Behavior_of_Reinforcement_Learning_Agents

- [8] A. Alharin, T.-N. Doan, and M. Sartipi, “Reinforcement Learning Interpretation Methods: A Survey,” *IEEE Access*, vol. 8, pp. 171058–171077, 2020, doi: <https://doi.org/10.1109/access.2020.3023394>.
- [9] D. Amodi et al., “Concrete Problems in AI Safety,” Jul. 2016. Available: <https://arxiv.org/pdf/1606.06565>
- [10] M. Speekenbrink and E. Konstantinidis, “Uncertainty and Exploration in a Restless Bandit Problem,” *Topics in Cognitive Science*, vol. 7, no. 2, pp. 351–367, Apr. 2015, doi: <https://doi.org/10.1111/tops.12145>.
- [11] E. Bender and A. Koller, “Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data,” *Association for Computational Linguistics*, 2020. Available: <https://aclanthology.org/2020.acl-main.463.pdf>
- [12] J. Banks, M. Olson, and D. Porter, “An experimental analysis of the bandit problem,” *Economic Theory*, vol. 10, no. 1, pp. 55–77, Jun. 1997, doi: <https://doi.org/10.1007/s001990050146>.
- [13] J. Dikes, V. Yazdanpanah, S. Stein. “Reinforced language models for sequential decision making.”. Arxiv, Available at: <https://arxiv.org/html/2508.10839v1>
- [14] E. Eigner, T. Handler. “Determinants of LLM-Assisted Decision Making”, Available at: <https://arxiv.org/pdf/2402.17385>
- [15] M. D. Lee, S. Zhang, M. Munro, and M. Steyvers, “Psychological models of human and optimal performance in bandit problems,” *Cognitive Systems Research*, vol. 12, no. 2, pp. 164–174, Jun. 2011, doi: <https://doi.org/10.1016/j.cogsys.2010.07.007>.
- [16] D. A. Berry, “A Bernoulli Twoarmed Bandit,” *The Annals of Mathematical Statistics*, vol. 43, no. 3, pp. 871–897, 1972, doi: <https://doi.org/10.2307/2240384>.
- [17] K. F. Vasilchenko and E. M. Chumakov, “Current status, challenges and future prospects in computational psychiatry: a narrative review,” *Consortium Psychiatricum*, vol. 4, no. 3, pp. 33–42, Sep. 2023, doi: 10.17816/cp11244.
- [18] “Rescorla-Wagner model of classical conditoning.” Available at: https://psych.hanover.edu/JavaTest/CLE/Cognition_js/RescorlaWagner/
- [19] M. J. Renner. “Learning the Rescorla-Wagner Model of Pavlovian Conditioning: An Interactive Simulation”, ResearchGate. Available at: https://www.researchgate.net/publication/240237597_Learning_the_Rescorla-Wagner_Model_of_Pavlovian_Conditioning_An_Interactive_Simulation

[20] R. Hendricks. “How do you interpret RMSE?”, deepchecks, Available at:
<https://www.deepchecks.com/question/how-do-you-interpret-rmse/#:~:text=A%20lower%20RMSE%20indicates%20that,that%20predictions%20are%20not%20accurate>.

[21] “Phenotype,” *Genome.gov*. <https://www.genome.gov/genetics-glossary/Phenotype>